

RiskPulse AI

AI/NLP Financial Risk Intelligence Engine

Strategic Portfolio Stress Testing — Module B

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2. Problem & Approach

Financial risk can emerge first as unstructured language in news and social-media discussion.

RiskPulse AI converts this text into structured sentiment, event and impact signals.

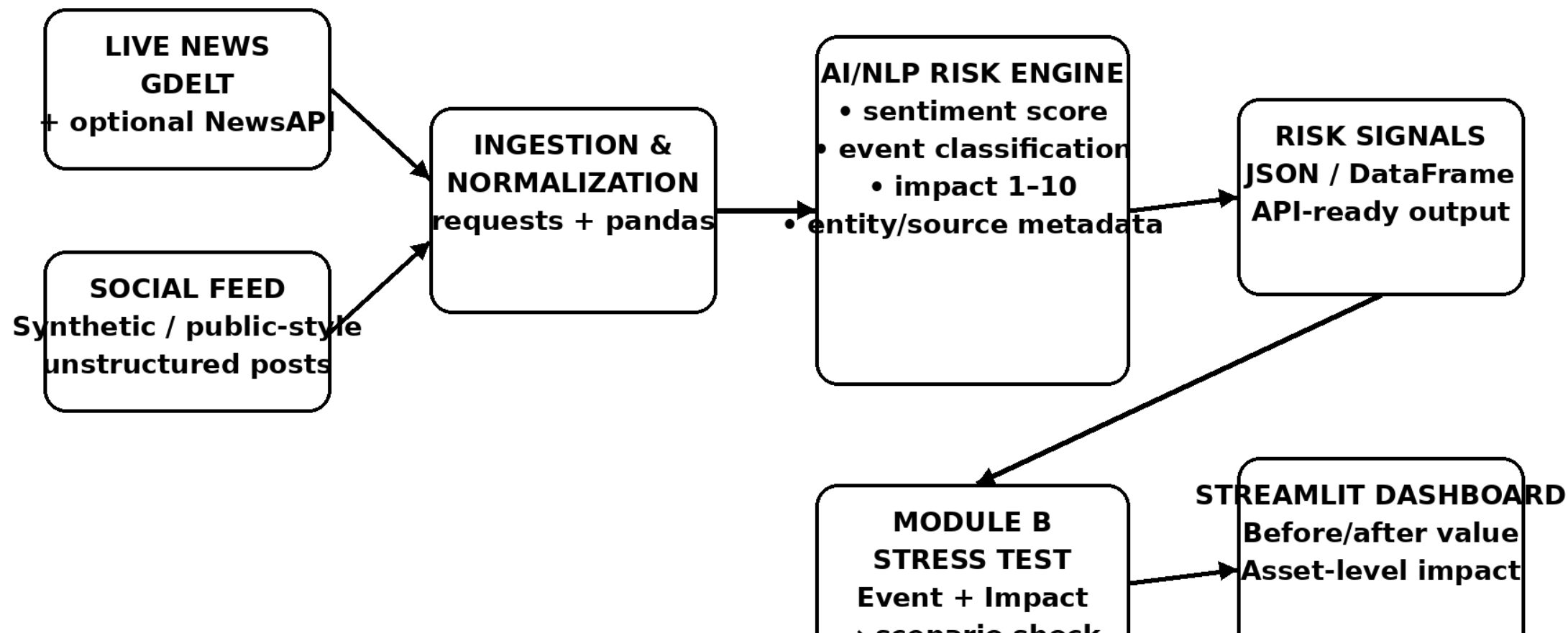
The downstream Module B consumes high-impact events and runs a transparent portfolio stress scenario.

The design prioritizes reproducibility, traceability and an end-to-end working prototype.

3. System Design

GDELT / optional NewsAPI → ingestion → hybrid NLP risk engine → structured risk signal → Module B stress test → Streamlit dashboard.

RiskPulse AI — Architecture & Data Flow



4. Implementation Highlights

Hybrid NLP: VADER sentiment + domain-specific event taxonomy + transparent impact scoring.

Event classes include Geopolitical, Macroeconomic, Credit Event, M&A, Regulatory, Product Launch and Earnings.

Synthetic portfolio covers equities, bonds, loans and derivatives.

FastAPI exposes the risk engine as a machine-readable endpoint.

Streamlit provides live ingestion, signal inspection and stress-test visualization.

5. Key Results

End-to-end pipeline converts unstructured text into structured risk signals.

Dashboard surfaces high-impact events and event distribution.

Stress test reports portfolio value before/after, simulated P&L and asset-level impact.

All synthetic demo data is included in the repository for reproducibility.

Scenario shocks are explicitly illustrative assumptions, not investment advice.

6. Domain Impact

Creates a standardized risk-intelligence layer between unstructured information and downstream risk workflows.

Provides traceability: source → text → classification → impact → portfolio effect.

Can support rapid scenario analysis when event severity crosses a configurable threshold.

Production path: financial event model, entity linking, streaming, confidence calibration, historical scenario calibration and monitoring.

7. Limitations & Next Steps

Current event/impact logic is a transparent hybrid MVP rather than a fully trained financial foundation model.

Synthetic social and portfolio data are used for reproducibility and confidentiality.

Production deployment would require authorized feeds, calibrated risk models, validation and governance.

Next: fine-tuned financial NLP, event deduplication, confidence scores, streaming infrastructure and model monitoring.